A Multi-Agent LLM Framework for Explainable Algorithmic Trading on NSE-Listed Equities

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***Abstract*—We present a multi-agent large language model (LLM) framework for explainable algorithmic trading on NSE-listed Indian equities, combining a peephole LSTM with temporal attention (PLSTM-TAL) for price-trend forecasting, LLM-scored news headline sentiment, and an adversarial bull/bear debate mechanism for signal reconciliation, gated by a deterministic, rule-based risk manager. The system is evaluated against classical technical baselines under a purged, embargoed walk-forward protocol and a complete Indian delivery-segment transaction-cost model (STT, stamp duty, exchange charges, GST, and slippage) over a 7.6-year out-of-sample window (2018–2026). The central empirical finding is negative: no architecture, agent configuration, or hyperparameter setting evaluated – across ten model architectures, three forecast horizons, two label constructions, a wider 40-name universe, and a 16-cell post-hoc hyperparameter sensitivity sweep – produces a Sharpe ratio whose 95% block-bootstrap confidence interval excludes zero after Holm-Bonferroni correction for multiple comparisons. A Monte Carlo power analysis indicates this window is statistically powered to detect a true Sharpe ratio of approximately 1.0 at 80% power, well above typical systematic-strategy performance, sharpening rather than resolving the null result. A regime-conditional analysis shows the risk overlay reduces drawdown by an order of magnitude during the 2020 COVID crash specifically, at the cost of forgone upside in calm regimes that outweighs this protection on average over the full window.**

I. Introduction

Algorithmic trading systems built on large language models have recently been proposed to combine the pattern-recognition strengths of deep sequence models with the natural-language reasoning of LLMs over unstructured information such as news headlines. This study reimplements and evaluates three influential approaches on Indian (NSE) equities under one evaluation protocol strict enough that a positive result, if found, would be trustworthy: a peephole LSTM with temporal attention (PLSTM-TAL) for price forecasting, an LLM headline-scoring methodology for news sentiment, and a multi-agent bull/bear debate framework for signal reconciliation and explainability.

II. Methodology

A. Technical Forecaster

A hand-implemented peephole LSTM cell with temporal attention (PLSTM-TAL) predicts the sign of the open-to-open forward return over a fixed horizon. Evaluation uses a purged, embargoed rolling walk-forward split (3-year train, 6-month test, an embargo exceeding the label horizon) to prevent label-construction leakage across fold boundaries.

B. LLM Headline Sentiment

Indian financial news headlines are scored by a constrained-label LLM classifier (Good / Bad / Unknown), following the share-price-directional prompting methodology of Lopez-Lira and Tang, run as a standalone event study before being used as a trading signal.

C. Bull/Bear Debate Framework

A bull and a bear case are scored per decision point via constrained-label LLM calls, contributing a residual conviction term to the combined signal; debate is gated to contested days only, based on cross-agent conviction disagreement.

D. Risk Management

Position sizing, volatility targeting, a portfolio-level gross-exposure cap, and a drawdown brake are implemented as deterministic, auditable rules rather than a further learned or LLM-driven component, since a risk constraint that can be argued out of firing is not a constraint.

E. Evaluation Protocol

All reported returns are net of a full Indian delivery-segment transaction cost model (STT, stamp duty, exchange charges, GST, and slippage). Statistical significance is assessed via block-bootstrap confidence intervals on the annualised Sharpe ratio, Holm-Bonferroni correction across the full set of configurations tried, and the Deflated Sharpe Ratio of Bailey and López de Prado, which explicitly penalises a track record for the number of configurations searched to find it.

III. Results

A. Architecture Ablation

TABLE I
Directional accuracy by architecture, purged walk-forward, 3 seeds, 16 folds.

Architecture	Accuracy	Std (3 seeds)	AUC	MCC	p vs. chance
LSTM	0.5045	0.0019	0.5049	0.0076	0.210
LSTM-TAL	0.5024	0.0005	0.5017	0.0056	0.506
PLSTM-TAL	0.4998	0.0048	0.5008	0.0007	0.958
PLSTM	0.4996	0.0003	0.4999	-0.0018	0.904

No architecture's out-of-sample directional accuracy is statistically distinguishable from chance (50%) at conventional significance levels.

TABLE II
GROSS VS. NET SHARPE RATIO, DELIVERY-SEGMENT INDIAN TRANSACTION COSTS.

Strategy	Sharpe (gross)	Sharpe (net)	Cost drag
Buy&Hold	+0.580	+0.578	+0.001
RSI(14)	+0.464	+0.424	+0.040
Tech+Regime	+0.577	+0.344	+0.233
Tech+Sent+Regime	+0.660	+0.177	+0.483
Full+Debate	+0.587	+0.131	+0.455
MACD	+0.466	+0.020	+0.446
KDJ+RSI	+0.234	-0.344	+0.578
Tech-only	-0.273	-0.412	+0.139
MeanReversion	+0.680	-1.408	+2.088
Tech+Sentiment	-0.114	-2.090	+1.975

B. Transaction-Cost Impact

Indian delivery-segment transaction costs (approximately 32 basis points round-trip at the measured turnover) materially erode every strategy's gross Sharpe ratio, in several cases reversing its sign.

C. Multi-Agent Ablation and the Deflated Sharpe Ratio

TABLE III
MULTI-AGENT ABLATION: BLOCK-BOOTSTRAP CONFIDENCE INTERVALS AND DEFLATED SHARPE RATIO (BAILEY & LÓPEZ DE PRADO), CORRECTING FOR THE MULTIPLE CONFIGURATIONS SEARCHED OVER THE COURSE OF THIS STUDY.

Configuration	Sharpe (net)	95% CI	$p(\text{Sharpe} > 0)$	Deflated Sharpe
Buy&Hold	+0.578	[-0.15, +1.34]	0.110	0.0168
RSI(14)	+0.424	[-0.33, +1.18]	0.250	0.0052
Tech+Regime	+0.344	[-0.34, +1.05]	0.324	0.0027
Tech+Sent+Regime	+0.177	[-0.51, +0.87]	0.605	0.0006
Full+Debate	+0.131	[-0.58, +0.85]	0.704	0.0004
MACD	+0.020	[-0.65, +0.65]	0.950	0.0001
KDJ+RSI	-0.344	[-0.97, +0.31]	0.336	0.0000
Tech-only	-0.412	[-1.25, +0.53]	0.443	0.0000
MeanReversion	-1.408	[-2.05, -0.77]	0.000	0.0000
Tech+Sentiment	-2.090	[-2.77, -1.40]	0.000	0.0000

Every configuration's Deflated Sharpe Ratio is substantially below the uncorrected $p(\text{Sharpe} > 0)$ figure, reflecting the number of configurations searched over the course of this study.

D. Hyperparameter Sensitivity

The null result is not an artefact of any single hand-set forecast horizon, transaction-cost assumption, or debate-escalation threshold: no cell in this pre-registered sensitivity sweep clears its own confidence interval above zero.

TABLE IV
HYPERPARAMETER SENSITIVITY SWEEP (A "+"-SHAPED DESIGN; SEE THE PRE-REGISTRATION IN THIS STUDY'S REPOSITORY). NO CELL'S INTERVAL EXCLUDES ZERO.

Configuration	Sharpe (net)	95% CI
horizon=1d, cost=20bps	+0.002	[-0.72, +0.78]
horizon=1d, cost=32bps	-0.051	[-0.77, +0.73]
horizon=1d, cost=50bps	-0.147	[-0.86, +0.62]
horizon=5d, cost=32bps	+0.101	[-0.58, +0.89]
horizon=20d, cost=32bps	+0.142	[-0.42, +0.88]
horizon=10d, cost=32bps	+0.092	[-0.64, +0.95]
conviction_floor=0.05	+0.073	[-0.67, +0.84]
conviction_floor=0.10	+0.066	[-0.67, +0.83]
conviction_floor=0.15	+0.080	[-0.65, +0.83]
conviction_floor=0.20	+0.151	[-0.57, +0.90]

IV. DISCUSSION AND CONCLUSION

Across ten model architectures, three forecast horizons, two label constructions, a wider 40-name universe, and a 16-cell hyperparameter sensitivity sweep, no configuration evaluated produces a statistically significant, cost-net positive Sharpe ratio on this dataset. A regime-conditional analysis indicates the deterministic risk overlay meaningfully reduces drawdown during an acute market crash, at a cost in forgone upside during calm periods that outweighs this protection on average over the full evaluation window. A Monte Carlo power analysis indicates the evaluation window is powered to detect a true Sharpe ratio of approximately 1.0 at 80% power – an unusually high bar relative to typical systematic-strategy performance – so this result should be read as "no edge above an unusually large threshold was found," not as a demonstration that no smaller edge exists. Future work should prioritise a materially more capable LLM backend for the sentiment and debate arms, a longer evaluation window, and point-in-time fundamental data.

REFERENCES

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